UNIVERSITY OF MAKATI

DEVELOPMENT OF QUIZARENA: A WEB-BASED GAMIFIED MULTIPLAYER EDUCATIONAL ONLINE QUIZ BATTLE SYSTEM USING RETRIEVAL-AUGMENTED GENERATION (RAG), ADAPTIVE RANDOMIZATION, AND LLM INTEGRATION

A thesis submitted to the faculty of College of Computing

and Information Sciences in candidacy for the Degree of

Bachelor of Science in Computer Science

(Application Development Elective Track)

DEPARTMENT OF COMPUTER SCIENCE

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In Partial Fulfillment

of the Requirements for the Degree

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

May 2026

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CHAPTER I

INTRODUCTION

The development of Artificial Intelligence (AI) has become a prominent technological advancement in the 21st century and has been widely considered the fourth industrial revolution, driven by the proliferation of big data and sophisticated computational architectures. According to Kuleto et al. (2024), AI algorithms and machine learning frameworks have become progressively significant in assisting human decision-making across a variety of disciplines, fundamentally reshaping traditional methodologies. Within the educational sector, the rapid evolution of educational technology has catalyzed a profound paradigm shift in contemporary pedagogical practices, predominantly through the integration of digital gamification and intelligent adaptive learning systems. Gamification, defined academically as the strategic transposition of game-design mechanics into non-game environments, has been empirically validated as a potent mechanism for amplifying student motivation, cognitive engagement, and longitudinal academic retention. As highlighted by Yang et al. (2023), the deployment of structured points, progression levels, and collaborative challenges transforms passive knowledge consumption into an interactive, goal-oriented endeavor. Furthermore, when gamified systems are deliberately aligned with social constructivist frameworks, they facilitate vicarious learning and collaborative knowledge construction, particularly in disciplines requiring critical analysis, deductive reasoning, and sustained attention.

Project Context

The rapid evolution of educational technology has catalyzed a profound paradigm shift in contemporary pedagogical practices, predominantly through the integration of digital gamification and intelligent adaptive learning systems. Gamified learning environments, when deliberately aligned with social constructivist frameworks, have been empirically validated as potent mechanisms for amplifying student motivation and facilitating collaborative knowledge construction. Concurrently, the integration of advanced Artificial Intelligence (AI) methodologies into these systems is fundamentally reshaping how student engagement and performance are monitored and sustained. The growing interest in highly interactive digital learning spaces and the rapid advancement of AI technologies have created an opportune intersection for research in this domain.

Within the broader domain of educational technology and artificial intelligence, the specific focus of this study revolves around the development of a gamified multiplayer educational platform tailored for complex, logic-heavy disciplines such as Mathematics and Computer Science. Previous research in gamified learning platforms has primarily concentrated on speed-based factual recall and rigid multiple-choice formats, lacking the specialized depth required for evaluating algorithmic logic or computational thinking. According to Baidoo-Anu and Ansah (2023), current educational technologies tend to operate in strict computational silos, presenting a clear knowledge gap in dynamic assessment.

This niche research area addresses the critical need to shift the design of educational platforms from hyper-competitive models to structured collaborative environments integrated with Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) for more accurate, context-specific cognitive assessment.

The core problem that this research endeavors to tackle is the development of an adaptive, AI-driven educational platform capable of assessing complex step-by-step reasoning rather than mere final answers. This proposed model aims to precisely capture, interpret, and evaluate the dynamic cognitive processes and written solutions of learners during individual and team-based assessments. By doing so, we aim to contribute to the broader field of educational technology by providing a tool that can automate complex academic grading, analyze the effectiveness of different problem-solving techniques, and assist in providing real-time, step-by-step feedback for students.

Existing research in educational technology showcases a distinct lack of tailored AI solutions for analyzing complex student inputs within collaborative platforms. Studies highlight the limitations of current systems; for instance, according to Chu et al. (2025), scaling and consistently capturing the nuances of open-text and complex responses remain significant challenges, limiting their widespread use in assessments compared to simple multiple-choice formats. Extant literature from Saldivar (2026) further indicates that poorly structured competitive environments often foster toxic behavioral dynamics and exacerbate performance disparities, as they disproportionately reward speed over deliberate reasoning. These limitations, alongside systemic educator concerns regarding heavy workloads and academic dishonesty, indicate a strong and substantial desire for a dedicated, AI-driven assessment model to reliably evaluate learning methodologies and performance.

The complexity of assessing subjects like advanced mathematics and programming, characterized by intricate multi-step problem-solving and algorithmic logic, presents challenges in traditional automated grading systems. The lack of specialized, collaborative evaluation models hinders progress in training quality, competitive fairness, and rigorous performance analysis. Consequently, learners may struggle to refine their critical thinking techniques, and instructors may face severe limitations in providing tailored, step-by-step feedback due to heavy administrative workloads. Addressing this problem through the integration of RAG and an LLM Solution Analyzer has the potential to significantly impact the effectiveness and accessibility of academic assessments, making them more engaging, fair, and conducive to deep learning for a broader audience.

Purpose and Description of the Study

In response to the challenges faced in accurately assessing complex problem-solving and fostering equitable academic collaboration, this research proposes the development of QuizArena, a specialized web-based, gamified multiplayer educational quiz battle system. This platform will employ state-of-the-art computer science techniques—including Retrieval-Augmented Generation (RAG) pipelines, Adaptive Randomization algorithms, and LLM-powered solution analyzers—to precisely track and interpret the dynamic cognitive processes exhibited in complex disciplines such as Mathematics and Computer Science. The core strategy of this solution involves the automated generation of high-quality, contextually accurate assessment items directly from instructor-provided syllabi using the RAG architecture, paired with robust editing interfaces for educators to manually validate AI-generated content. Key system features include flexible multiplayer modes, an LLM-driven Solution Analyzer capable of evaluating handwritten mathematical inputs step-by-step, and an algorithm evaluator designed to assess programming logic rather than just final outputs.

The primary purpose of this study is to design, develop, deploy, and rigorously evaluate the QuizArena platform, tailored explicitly to the distinctive evaluation requirements of logic-heavy higher education subjects. This purpose directly aligns with the urgent need to address the pedagogical limitations of existing hyper-competitive, multiple-choice platforms when applied to computational and mathematical domains. By achieving this purpose, the study aims to provide students, instructors, and educational institutions with a valuable tool capable of automating complex academic assessments, mitigating speed-induced cognitive anxiety, offering flexible collaborative testing modes, and ultimately ensuring a more equitable and profound learning experience.

This study holds significant implications both within the realm of educational technology and the broader field of artificial intelligence integration. The proposed QuizArena model has the potential to revolutionize traditional classroom assessments by offering real-time, precise, and context-specific evaluations while significantly reducing instructor workloads through automated, editable content generation. Moreover, it contributes practically to the field of computer science by addressing the technical challenges posed by real-time multiplayer state synchronization and dynamic team balancing using covariate-adaptive matching algorithms. Ultimately, this research seeks to bridge the gap between interactive gamified engagement and rigorous algorithmic assessment, advancing the efficacy of digital education ecosystems and setting a new standard for AI-assisted pedagogical tools.

Objectives

The general objective of this study is to design and develop QuizArena, a web-based gamified multiplayer educational online quiz battle system to facilitate balanced academic competition and collaborative reasoning across English, Mathematics, and Coding from course syllabi, handwritten sketches, and programming inputs using Retrieval-Augmented Generation (RAG) pipelines, Adaptive Randomization, and Large Language Model (LLM) integrations, which may automate academic assessments, mitigate speed-induced cognitive anxiety, and enhance collaborative learning to make the educational experience convenient and engaging for both students and instructors.

Specific Objectives

In order to fulfill the main objective of this project, the authors constructed the following specific objectives:

1. Examine and understand the fundamental pedagogical mechanics of flexible multiplayer gamified learning, team-based bracketing, and the collaborative reasoning methodologies used in assessing diverse academic subjects like English, Mathematics, and Coding.

2. Explore existing gamified educational platforms and artificial intelligence frameworks relevant to the development of automated retrieval-augmented question generation, visual sketch analysis, and automated code evaluation systems.

3. Analyze, understand, and perform preprocessing activities on unstructured instructional materials, curriculum syllabi, and student-submitted mathematical and programming data gathered toward the development of the RAG pipeline and automated grading algorithms.

4. Train and evaluate the RAG pipeline, Adaptive Randomization algorithms, and LLM-driven Solution Analyzers using the following evaluation metrics:

a. Retrieval-Augmented Generation Assessment Scale (RAGAS) for Contextual Precision and Answer Relevancy; and

b. Demographic Parity and Bounded Individual Loss for competitive matchmaking and team fairness calibration.

5. Design and develop a web-based application with the following key features:

   a. Automates the generation of lesson-aligned assessment items using a professor-validated RAG pipeline;

   b. Provides manual editing tools for educators to validate, modify, and refine AI-generated questions;

   c. Features an LLM-powered Solution Analyzer to evaluate handwritten or sketched mathematical step-by-step solutions and programming logic;

   d. Features a real-time, flexible multiplayer quiz system synchronized via WebSockets supporting individual seatwork, small teams, and Battle Royale formats;

   e. Allows dual scoring modes (Speed Mode and Discussion Mode) with a designated leader tie-breaking mechanic to resolve voting stalemates in collaborative play;

   f. Seeding and bracketing system utilizing the Adaptive Randomization algorithm for skill-based team matchmaking;

   g. Utilization of a comprehensive Professor Dashboard in order to visualize real-time match analytics and sort results (e.g., by accuracy or alphabetically);

h. Managing section registration, student join requests, and allowing section reassignment of student per teacher for student filtering;

i. Provides an adaptable, responsive view of the web-application on both desktop and mobile devices;

j. Incorporates misconception-type tagging within the Solution Analyzer feedback, classifying each incorrect response into a specific error category (e.g., sign error, off-by-one error, syntax fault, or logic fault) and aggregating error frequency per class to support diagnostic, rather than purely evaluative, feedback;

k. Displays a source citation for every RAG-generated question, referencing the specific syllabus chunk or page from which the item was retrieved, to support explainable and auditable content generation;

6. Test the functionality (based on test cases) and non-functionality of the system:

   a. Functional Requirements: AI question generation accuracy, solution analyzer accuracy, multiplayer state synchronization, and teacher editing tools;

   b. Non-Functional Requirements: Performance Efficiency, throughput, load testing, and stress testing;

7. Evaluate the system using a set of metrics based on the ISO/IEC 25010

standards which focuses on:

a. Functional Suitability;

b. Performance Efficiency;

c. Usability; and

d. Reliability.

8. Implement and deploy the system.

Scope and Limitations

The primary focus of this research is the utilization of the algorithm for the design and development of an application. The algorithm serves as the foundational technology driving the core functionalities of the application. The core focus of this research revolves around the application of Retrieval-Augmented Generation (RAG), Adaptive Randomization, and LLM Solution Analyzer algorithms, specifically adapted for parsing instructional materials to autonomously generate questions, evaluating handwritten math sketches or programming code, and dynamically calibrating team brackets. These algorithms form the backbone of the application's automated assessment and equitable matchmaking capabilities.

The study discusses system features that the application incorporates to address its intended objectives. While not detailed in this section, these features guide the application's functionality and user experience. The study outlines a comprehensive list of system features, which form the functional framework of the application. These features include a flexible multiplayer battle system, manual teacher editing interfaces for AI-generated content, dual scoring modes (Speed and Discussion Mode), a designated leader tie-breaker, and a Professor Dashboard for real-time analytics. While not detailed in this section, they guide the overall functionality and user experience of the application.

The geographical scope of this research extends to the University of Makati, targeting the student and faculty population within the College of Innovative Teachers Education. Data collection, testing, and user interaction are expected to occur within this defined geographical region.

The study encompasses data collected and system development activities conducted during the period from May 2026 to May 2027. It acknowledges that technological advancements beyond this temporal scope may not be considered in the study.

Tools for System Design and Development: The study employs a range of tools and technologies for the design and development of the application. These tools include:

a. Next.js, a react-based web application framework, as the primary web development tool;

b. Python v3.x as the language for writing the backend AI microservices;

c. PostgreSQL database engine as the local data storage for the web application;

d. Prisma ORM as a tool to manage PostgreSQL queries;

e. LangChain for structuring the RAG pipeline and LLM capabilities;

f. Socket.IO and Redis which handles the low-latency real-time state synchronization of the multiplayer environment;

g. Tailwind CSS which handles the design of the web-application's user interface and its page responsiveness;

Limitations

Despite the study's objectives, there are certain limitations that should be acknowledged:

This study heavily relies on the specified algorithm for its success. The effectiveness of the application is contingent on the performance and accuracy of this algorithm. Any limitations or constraints of the algorithm may consequently impact the application's performance. This study is heavily reliant on the RAG and LLM algorithms for its effectiveness. The accuracy and performance of the application are intrinsically linked to the capabilities and limitations of these algorithms. Any constraints or inaccuracies in the algorithms, such as retrieval hallucinations or constraints from highly illegible handwritten sketches, may impact the application's performance and mandate human oversight via professor editing tools.

The findings and outcomes of this research are specific to the context, geographical area, and time frame defined in the scope. The generalizability of the application's performance to other regions or periods may be limited. The findings and outcomes of this research pertain specifically to the context, geographical area, and time frame defined within the study's scope. The application's performance may not be directly generalizable to other regions, non-logic-heavy subjects, or time periods without further validation.

External factors such as network connectivity, hardware constraints, and user-specific behaviors are beyond the study's control and may influence the application's performance. External factors, including network connectivity, hardware limitations, and variations in user behaviors, are beyond the study's control but may influence the application's real-world performance, especially as real-time mechanics like live chat and synchronized scoring are highly sensitive to network latency.

Definition of Terms

This specific section of the paper defines significant terminologies that were utilized in order to fully comprehend the various concepts of the overall project.

In the design of our system, *gamification* refers to the strategic transposition of game-design mechanics, such as points, leaderboards, and collaborative challenges, into non-game educational environments to amplify student motivation, cognitive engagement, and longitudinal academic retention. The core engine for dynamic assessment relies on *retrieval-augmented generation*, which is defined as an advanced artificial intelligence architecture that merges the generative capabilities of language models with the dynamic retrieval of external, verified knowledge repositories to ensure contextually accurate outputs and mitigate algorithmic hallucination. Operating as the primary cognitive agent for grading, *large language models* represent sophisticated machine learning frameworks trained on vast datasets used within this system to evaluate logic, synthesize text, and act as the core reasoning engine for automated assessment. To ensure competitive fairness during multiplayer matches, *adaptive randomization* denotes the covariate-adaptive matchmaking algorithm used to dynamically balance teams based on historical diagnostic accuracy and performance covariates. For evaluating step-by-step problem-solving, the *solution analyzer* is an integrated, vision-capable language model module designed to evaluate handwritten mathematical sketches and programming code logic, moving beyond binary multiple-choice parameters. Within the multiplayer platform, players can access the *battle royale mode*, a specific game configuration where individuals or teams compete simultaneously in real-time until a single victor or surviving team remains. In collaborative team play, voting deadlocks are resolved using the *designated leader tie-breaker*, a game mechanic that grants a specified team member the final executive decision to progress through the quiz battle. The fairness and equity of team balancing are quantitatively measured using *demographic parity*, a fairness calibration metric utilized to ensure matchmaking does not exhibit bias based on predefined demographic or performance variables. Furthermore, the accuracy of our question generation pipeline is evaluated using *contextual precision*, a metric under the Retrieval-Augmented Generation Assessment Scale that measures the exactness and relevancy of retrieved instructional documents against the generated question. To provide diagnostic, rather than purely evaluative, insight, the Solution Analyzer applies *misconception tagging*, the classification of an incorrect student response into a specific, named error category (e.g., sign error, off-by-one error, syntax fault, or logic fault) so that error patterns can be aggregated and surfaced to instructors as diagnostic, class-level insight. Complementing this, *source attribution* denotes the retention of a traceable citation, linking each RAG-generated assessment item back to the specific syllabus chunk or page from which it was retrieved, allowing instructors to audit and verify the grounding of AI-generated content. Finally, to maintain immediate state updates across clients, the system utilizes *websockets*, a full-duplex communication protocol designed to maintain persistent, low-latency connections between the client and the server, enabling real-time state synchronization for the collaborative quiz environment.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

The transition of contemporary higher education towards highly interactive, digital, and intelligent ecosystems has necessitated the development of advanced pedagogical systems that synthesize gamified mechanics with artificial intelligence. Gamification, defined as the strategic application of game design elements like points, levels, and badges in non-game environments, has gained significant traction for its capacity to transform routine instruction into active, motivating, and cognitively engaging experiences. Traditional physical instruction, which relied primarily on manual assessment, has switch-mapped into virtual and online environments where the need to automate visual and logical perception is paramount. Historically, early online quiz systems operated strictly as static multiple-choice questionnaires, focusing heavily on speed-based recall and individualistic competition, which frequently induced cognitive anxiety and performance disparities. According to Tapales (2023), introducing gamified elements within classroom learning environments paves the way to a more conducive learning environment by significantly increasing attention spans and promoting active knowledge construction. However, as the educational domain switches to advanced, logic-heavy subjects like Mathematics and Computer Science, traditional quiz structures fail to evaluate the deep, multi-step cognitive reasoning of learners.

To address these assessment challenges and automate the grading of complex reasoning, research has transitioned from rule-based scoring systems to sophisticated artificial intelligence models, specifically utilizing Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs). According to Vatankhah Barenji et al. (2026), RAG systems integrate a large language model with a structured retrieval mechanism to access rubrics, exemplar responses, and verified course syllabi, producing highly reliable and contextually grounded feedback at scale. This technical integration mitigates the prominent hallucination issues of standalone LLMs and bridges the gap between automated scaling and qualitative evaluation. As a result, other RAG architectures, multi-agent grading frameworks, and adaptive matchmaking algorithms must be examined chronologically to establish the theoretical and scientific foundations of this study, positioning the proposed QuizArena platform within the broader academic discourse of digital transformation.

Chronological Review of Scholarly Works (2023 - 2026)

The exploration of educational technology and AI integration over the last four years demonstrates a rapid evolution from initial software acceptance studies to highly complex, agentic, and multimodal systems.

Early Developments and Foundations (2023)

In the wake of the public release of generative conversational models, early research in 2023 focused heavily on exploring basic capabilities, pedagogical theoretical frameworks, and immediate academic integrity implications. Baidoo-Anu and Ansah (2023) synthesized literature to outline the potential benefits of generative models, highlighting the automated generation of prompts for formative assessments and personalized explanations as transformative features, while cautioning against superficial evaluation. Simultaneously, Dwivedi and Kshetri (2023) provided a multidisciplinary perspective on opportunities and challenges, arguing that while productivity increases, academic institutions must establish robust anti-gaming and authorship verification safeguards. To safeguard learning, Kung et al. (2023) benchmarked LLMs against standardized licensing exams, indicating that while base models perform adequately, they remain highly prone to fabricating citations if not anchored by external databases. This limitation was theoretically addressed by Lewis et al. (2023), who formalized the foundational principles of Retrieval-Augmented Generation (RAG), proving that combining dense passage retrievers with sequence-to-sequence generators drastically reduces semantic hallucinations and ensures factual grounding. In parallel, work on model transparency began to address the reliability of AI-generated content: Gao et al. (2023) introduced ALCE, a foundational framework enabling large language models to generate text with verifiable in-line citations, establishing citation precision and recall as standard metrics for auditable AI outputs. In the assessment domain specifically, Gorgun and Botelho (2023) applied natural language processing and clustering techniques to automatically identify common student math misconceptions from both closed- and open-ended responses, demonstrating that error classification, not just correctness scoring, could be automated at scale.

Concurrently, research on gamified frameworks validated their positive influence on student psychology and retention. Yang et al. (2023) implemented the Gamified Artificial Intelligence Educational Robot (GAIER) system, utilizing a theory-driven "GAFCC" model to prove that strategically structured challenges and feedback loops significantly enhance cognitive flow and active participation in technical courses. Locally, Tapales (2023) developed the Reinforcement-Assessment Development Theory, formulating five core axioms and ten propositions which mathematically demonstrated that gamified elements employed in classroom discussions extend learners' attention span and accelerate their problem-solving capacity. However, Serafica and Muria (2023) investigated academic burnout and resilience, warning that unoptimized digital workloads lead to high student disengagement, reinforcing the need for adaptive systems that calibrate task difficulty.

Technical Advancements and Systemic Integrations (2024)

By 2024, the research landscape transitioned from theoretical explorations to the development of structured, domain-specific systems. In the Philippines, Villarino (2024) conducted a mixed-methods survey in a Cebu-based rural higher education institution, revealing that 78.54% of students had adopted tools like ChatGPT but expressed critical anxieties regarding biased information, the erosion of critical thinking, and the absence of institutional policies. Leonillo (2024) supported these findings in Bacolod City, showing that while AI integration somewhat enhanced teacher efficiency (M = 3.60), actual usage was hindered by limited training, suggesting the necessity of intuitive platforms with automated oversight tools.

To resolve the limitations of standalone competitive game designs, Salcedo (2024) executed a systematic mapping study, identifying that leaderboards, when used as isolated extrinsic motivators, often induce performance anxiety and subsequent disengagement once reward thresholds are met. To counter this, Duterte (2024) implemented collaborative challenges alongside individual points in Manila universities, proving that combining cooperative play with personal metrics yields statistically significant improvements in motivation.

In mathematics and coding, the necessity of assessing logical pathways was actively pursued. Borbe and Limpiada (2024) implemented gamified algebra lessons in Quezon, recording a grand mean score jump from 1.89 to 5.51, proving that interactive, game-based quizzing bridges conceptual understanding gaps in numerical reasoning. From a computer science perspective, Mullins et al. (2024) evaluated the technical parameters of RAG systems in education, demonstrating that optimizing document chunk sizes and embedding models directly influences the generation correctness of assessment items, achieving

RAGAS evaluation scores of over 50% for context grounding. This was technically enhanced by Feng et al. (2024) and Latif and Zhai (2024), who proved that fine-tuning language models on structured, rubric-based criteria allows automated scorers to align tightly with human-assigned grades. Addressing the diagnostic quality of AI feedback, Scarlatos et al. (2024) proposed a five-component rubric for evaluating math feedback, explicitly scoring whether generated feedback correctly identified the student's underlying misconception, then applied reinforcement learning to optimize feedback generation against that rubric. Concurrently, Huang et al. (2024) advanced citation-generation research by training language models to produce in-text citations via fine-grained rewards, improving the correctness and verifiability of claims attributed to retrieved source documents.

Multi-Agent Architectures and Real-Time Systems (2025)

The year 2025 was characterized by the emergence of multi-agent AI systems, multimodal vision-language parsing, and the localization of these tools within the Philippine curriculum. Bayoneta, Posecion, and Trayco (2025) conducted a qualitative analysis of gamification in local mathematics instruction, confirming that when game elements are constructively aligned with the ADDIE design model, students experience a 15% average improvement in short-term retention and 10% gains in delayed recall. Fodale (2025) synthesized these trends, noting a massive transition towards integrating gamification with mobile learning and adaptive personalization to promote equity. Locally, Santiago (2025) explored ChatGPT-assisted collaborative action research in environmental science classes, demonstrating that generative strategies successfully prompt higher-order thinking when structured within the iterative "IDEE" framework.

However, the limits of ungrounded models were heavily documented by Serdenia (2025) and Cabuquin (2025), who found that while Filipino students moderately utilized conversational models for research, they frequently encountered redundant, inadequate, and inaccurate information, illustrating the urgent need to deploy localized RAG architectures in educational apps.

To resolve these factual inaccuracies at the model level, Qiu et al. (2025) developed "SteLLA," a structured grading system using RAG to extract rubric-defined knowledge points before scoring short answers, achieving substantial agreement with human graders. Concurrently, Chu et al. (2025) launched "GradeOpt," which optimizes grading prompts dynamically using a multi-agent framework, proving that cooperative agent networks can consistently capture instructional nuances at scale. To handle multi-step mathematical logic, Bachmann and Nagarajan (2025) introduced the Latent Diffusion Reasoner (LaDiR), which maps text reasoning steps into expressively encoded blocks to overcome the autoregressive limits of single-token generation. This was supported by Gierse et al. (2025) and Zhang and Zhao (2025), who demonstrated that RAG pipelines utilizing localized textbook databases achieved up to 97.2% factual precision on technical queries. Extending misconception detection into programming education, Oli et al. (2025) evaluated zero-shot prompting, supervised fine-tuning, and preference alignment of LLMs to identify gaps and misconceptions in students' self-explanations of code, finding that fine-tuned models most reliably distinguished genuine conceptual gaps from superficial errors. In the same year, Abdul Wahid et al. (2025) validated a curriculum-grounded RAG pipeline for generating secondary mathematics multiple-choice questions, introducing a RAG-based question-answering verification step that confirmed each generated item could be traced back to and answered from the source curriculum document.

Autonomous Agents and Scaled Implementations (2026)

The current year, 2026, marks the maturity of autonomous, agentic systems capable of handling real-world educational deployment. Locally, Angara (2026) issued national foundational guidelines on AI integration, driving reforms through programs like AGAP.AI and DepEd's E-CAIR (Education Center for AI Research) to train millions of Filipinos in AI literacy and ethical use. Saldivar (2026) conducted a meta-synthesis of AI-driven gamification in higher education, mapping out how teachers' roles have evolved from simple instruction to active interpretation of data-driven learning analytics. This is practically illustrated by Porras et al. (2026) with the deployment of "LET do IT," an online reviewer that integrates Agile development with rule-based performance analytics, enabling self-regulated study pathways.

To address high-stakes assessments, Vatankhah Barenji et al. (2026) validated an LLM-powered RAG assessment framework, achieving an extraordinary 94% to 99% grading agreement with human evaluators by strictly restricting generative output to validated marking schemes. Cong et al. (2026) introduced an Item Response Theory (IRT) evaluation model for automated short-answer grading, mapping out model performance across a spectrum of student response difficulties, which highlighted the tendency of LLMs to struggle with partially correct, incomplete reasoning.

To automate the evaluation of handwritten mathematics, CalcTutor (2026) implemented a multi-agent pipeline featuring a LaTeX Agent for optical transcription, a Solver Agent for logical mapping, and a Grader Agent for rubric-based grading, achieving a weighted F1-score of 0.934. Finally, Kleiman et al. (2026) introduced "ArguAgent," a system that utilizes stance clustering and quality scoring to automatically group students in real-time, achieving a 3.2-fold improvement over random grouping in fostering inclusive, peer-led argumentation in STEM classrooms. Most recently, Liu et al. (2026) proposed a cognitive-uncertainty-guided knowledge distillation framework for classifying student misconceptions, achieving a 17.8% improvement in top-3 diagnostic accuracy while requiring only a small, deployable model, underscoring that fine-grained misconception classification is now feasible even under real-time application constraints. In parallel, Xu et al. (2026) deployed a curriculum-grounded LLM-as-Judge marking pipeline that assembles verifiable, authorized syllabus artefacts before every judgment, yielding marking outcomes comparable to human tutors while producing justifications that are transparently traceable to source material.

Synthesis

The comprehensive synthesis of the reviewed literature from 2023 to 2026 reveals a diverse array of studies that have contributed to the understanding of AI-driven assessment and gamified learning. A range of methodological approaches has been employed in these studies, offering multifaceted insights into the technical, cognitive, and sociocultural dimensions of educational technology. Early investigations in the post-pandemic era primarily focused on qualitative analyses, utilizing phenomenological approaches, expert observations, and semi-structured interviews to document initial user perceptions and ethical anxieties surrounding generative AI. As the field progressed, a distinct shift towards quantitative and experimental methods became evident. Controlled quasi-experiments emerged, utilizing pretest-posttest designs to quantify the precise impacts of gamified lessons on student engagement, motivation, and conceptual retention in logic-heavy subjects. Concurrently, data scientists and software developers adopted technical benchmarking methodologies, applying evaluation metrics such as RAGAS, Item Response Theory, and inter-grader agreement rates to measure the accuracy and robustness of RAG pipelines and multi-agent grading frameworks. The diversity in methodological choices—encompassing qualitative thematic analysis, document-based qualitative ADDIE evaluations, and quantitative correlational research—underlines the multidimensional nature of AI-integrated gamification, reflecting both its rigorous technological foundations and its profound pedagogical implications.

Despite the wealth of research on educational technology, several critical research gaps have come to the forefront of the existing literature. Firstly, there is a notable geographic bias, with the vast majority of sophisticated RAG and multi-agent AI assessment systems predominantly developed and tested within Western or highly industrialized Asian contexts. This geographic limitation impedes the broader understanding of AI-driven tools in developing regions, particularly in the Philippines where unequal digital infrastructure and the urban-rural divide constrain technological adoption. Secondly, while gamification is widely validated, the temporal scope of local studies often falls short of conducting longitudinal assessments, leaving its long-term effects on conceptual retention and academic performance underrepresented. Thirdly, an underexplored area relates to the integration of visual, step-by-step logic assessment within multiplayer gamified systems. Traditional platforms remain heavily reliant on static multiple-choice formats, completely failing to capture the rich, multi-step problem-solving processes required in advanced mathematics and coding. Fourthly, even among the AI-driven assessment tools reviewed, feedback and content generation are largely treated as black-box outputs: grading systems typically report whether a response is correct without classifying the specific misconception behind an error, and RAG-based question generators rarely expose which source material a generated item was retrieved from, limiting instructors' ability to diagnose learning gaps or audit AI-generated content. In light of these research gaps, the current study assumes a crucial role in addressing these limitations. By developing QuizArena, this research proposes a web-based, gamified multiplayer platform specifically tailored to the Philippine higher education context, integrating a RAG pipeline for localized curriculum alignment with per-question source attribution, an LLM-driven Solution Analyzer for evaluating multi-step handwritten sketches and coding logic with misconception-type tagging, and covariate-adaptive adaptive randomization to ensure balanced and fair competition.

Table 1. The following is the list of the existing foreign literature on Artificial Intelligence, RAG, and Automated Assessment.

Paper / Area of Focus	**Proponents**	**Algorithm / Approach**	**Accuracy / Metric / Outcome**
LLM-Powered Assessment in Higher Education	Vatankhah Barenji et al. (2026)	RAG + structured prompt engineering	94%–99% grading agreement with human evaluators

SteLLA: Structured Grading with RAG	Qiu et al. (2025)	RAG-grounded short answer evaluation	Substantial agreement; highly precise factual capture
GradeOpt: Multi-Agent Grading	Chu et al. (2025)	Multi-agent guidelines optimization	Human-level conceptual nuance capture
IRT Evaluation for LLM Grading	Cong et al. (2026)	Item Response Theory grading model	Response-level difficulty and robustness mapping
CalcTutor: Multi-Agent Math Grading	CalcTutor (2026)	Multi-agent (LaTeX, Solver, Grader)	0.931 weighted agreement; 0.934 F1-score
ArguAgent: Real-Time Stance	Kleiman et al. (2026)	Semantic clustering + grouping algorithm	95.4% successful grouping; 3.2x over random

Grouping			
Fostering AI Literacy via Games	Ng et al. (2024)	Gamified e-books (TreasureIsland)	Significant gain in motivation and AI self-efficacy
RAG Chunk Evaluation in Education	Mullins et al. (2024)	Dense passage retrieval chunk evaluation	Context grounding correctness evaluation

The eight foreign studies presented in Table 1 collectively demonstrate the rapid maturation of AI-powered assessment and automated grading systems from 2024 to 2026. Taken together, these works establish a clear technological trajectory: from foundational RAG-based short answer scoring (Qiu et al., 2025; Mullins et al., 2024) toward more sophisticated multi-agent, multimodal pipelines capable of evaluating handwritten mathematical logic and facilitating real-time student grouping (CalcTutor, 2026; Kleiman et al., 2026). Vatankhah Barenji et al. (2026) and Chu et al. (2025) specifically validate that high inter-rater agreement with human evaluators is achievable when large language models are grounded in structured retrieval mechanisms, directly supporting the design rationale of QuizArena's RAG pipeline. The convergence of these studies toward Item Response Theory integration (Cong et al., 2026) and gamification-augmented AI literacy (Ng et al., 2024) further underscores the imperative for a system that not only automates assessment but also delivers a motivationally engaging and psychometrically sound evaluation experience—precisely the dual mandate that QuizArena is engineered to fulfill.

Table 2. The following is the list of the gathered local literature about Digitalization and Gamification in the Philippine Education System.

Title	Proponents	Description
Rethinking Learning in Higher Education Through AI-Driven Gamification: A Meta-Synthesis	Saldivar (2026)	Evaluated educators' evolving roles and experiences across 22 qualitative studies utilizing Publish or Perish.
Gamified Teaching-Learning Strategies and Instructional Practices in Mathematics: Effects on Student Engagement and Learning Retention	Bayoneta, Posecion, & Trayco (2025)	Document-based ADDIE qualitative evaluation showing a 15% average gain in short-term math retention.

LET Do IT: An Online Licensure Examination for Teachers Reviewer with Performance Analytics	Porras et al. (2026)	Web-based online LET reviewer developed via Agile-Scrum integrating transparent rule-based analytics.
Gamification in Language Education: Reinforcement-Assessment Development Theory	Tapales (2023)	Mixed-methods study generating 5 core axioms and 10 propositions regarding gamified attention span.
The Impact of Educational Gamification on Student Learning Outcomes	Duterte (2024)	Mixed-methods quasi-experiment with 133 undergraduates in Manila showing significant motivation gains.
Effect of Gamification in Improving the Performance in Mathematics of Grade 7 Learners	Borbe & Limpiada (2024)	Quasi-experimental pretest-posttest showing mathematics MPS gains from 28.69% to 82.75% via Kahoot.
Adoption, perceptions, and ethical implications	Villarino (2024)	Sequential explanatory mixed-methods study of

of AI tools among rural Philippine college students		451 students in Cebu; 78.54% reported ChatGPT adoption.
AI-Assisted Education: An Investigation into its Perceived Effects on Teacher Efficiency and Student Performance	Leonillo (2024)	Correlational study of 86 teachers and 274 students in Bacolod; reported moderate (3.60) AI efficiency.

The eight local Philippine studies presented in Table 2 collectively illuminate both the growing enthusiasm for and systemic barriers to digital gamification and AI integration in Philippine higher education. Across the reviewed works, a consistent finding emerges: gamified instructional strategies—whether implemented through Kahoot-based quizzes (Borbe & Limpiada, 2024), collaborative game mechanics (Duterte, 2024), or theoretically grounded reinforcement frameworks (Tapales, 2023)—yield measurable improvements in student engagement, retention, and academic performance. However, Villarino (2024) and Leonillo (2024) jointly reveal that adoption is constrained by insufficient institutional training and anxieties over AI reliability, signaling that any domestically deployed platform must prioritize intuitive design and transparent oversight mechanisms. At the systems level, Saldivar (2026) and Porras et al. (2026) demonstrate that both meta-level analyses and practical deployments of AI-driven tools are increasingly viable within the Philippine context, provided they account for localized pedagogical standards and institutional readiness. These findings collectively affirm the contextual relevance and practical urgency of QuizArena’s development as a locally grounded, AI-augmented educational platform designed specifically for the assessment complexities of Philippine higher education.

Conceptual Model of the Study

The conceptual model serves as a visual representation of the key components, variables, and relationships involved in the QuizArena research. It provides a roadmap for understanding the flow of information, processes, and interactions within the study. In this section, we discuss the key elements of the conceptual model and how they contribute to achieving the research objectives.

The Input-Process-Output (IPO) diagram is a valuable tool for illustrating the flow of information and activities in the QuizArena research. It consists of three main components: Inputs, which represent the data and variables entering the research process; Processes, which encompass the activities and procedures applied to transform these inputs; and Outputs, which are the finalized results, features, and outcomes generated by the study.

INPUTS	PROCESSES	OUTPUTS
1. Unstructured syllabus	1. Semantic chunking &	1. Verified,

PDFs & instructional text documents. 2. Student diagnostic data, skill ratings, & accuracy logs. 3. Handwritten mathematical steps & structured programming code. 4. WebSocket packet transmissions and user manual override configs.	embedding. 2. Prompt-engineered RAG question gen. 3. LLM visual OCR & code logic analysis. 4. Dynamic covariate-adaptive matching. 5. Websocket state & room synchronization. 6. Manual professor validation/editing. 7. Tie-breaking designated leader logic. 8. ISO/IEC 25010 standard evaluations.	lesson-aligned quiz and assessment database. 2. Graded step-by-step solutions & personalized student feedback. 3. Synchronized live battle score rankings and team brackets. 4. Analytic Professor Dashboards & comprehensive engagement logs.

In this study, inputs consist of various data sources that serve as the foundational data for analysis and development. The primary inputs include unstructured instructional materials such as PDF syllabi, lecture presentations, and text documents provided by educators to construct the question bank knowledge base. Additionally, inputs encompass student diagnostic profiles, historical accuracy data, and skill ratings used to calibrate matchmaking. For assessment evaluations, inputs consist of student-submitted handwritten mathematical solutions (uploaded as images) and structured programming code snippets. Finally, real-time client-side WebSocket packet transmissions (such as answers, team votes, and chat messages) and manual system configuration inputs (such as teacher overrides and question edits) enter the system as dynamic variables.

The processes in this research involve a series of interconnected technical and analytical steps designed to transform raw inputs into a synchronized, intelligent application. First, data preprocessing activities are applied to chunk and embed unstructured documents into a high-speed database. Second, semantic search and question generation are executed via the RAG pipeline. Third, manual educator validation and editing processes allow professors to overwrite and refine AI-generated questions. Fourth, the LLM-driven Solution Analyzer processes handwritten mathematical steps using vision-based OCR and compares programming structures against logical execution models. Fifth, covariate-adaptive matchmaking is executed via the Adaptive Randomization algorithm to dynamically balance team weights. Sixth, multiplayer game states are synchronized across clients using low-latency WebSockets, resolving deadlocks through designated leader tie-breaking processes. Collectively, these iterative software engineering and data science processes are executed within a CRISP-DM-AGILE framework to develop the comprehensive QuizArena system.

The research outputs comprise various functional, analytical, and evaluative components that reflect the study's overall contributions. The primary output is the fully deployed, web-based QuizArena application, which automates the evaluation and scoring of diverse academic subjects (English, Mathematics, and Coding) in real-time. Additionally, the study yields verified, curriculum-aligned question repositories and step-by-step diagnostic feedback files generated by the Solution Analyzer. For educators, the outputs include an interactive, sortable Professor Dashboard providing real-time match analytics and comprehensive student engagement reports. For the academic community, the outputs culminate in a thoroughly evaluated system tested under ISO/IEC 25010 standards, offering valuable insights into the scalable integration of generative AI and collaborative gamification in higher education.

CHAPTER III

DESIGN AND METHODOLOGY

This chapter presents the research design and methodology employed in the development and evaluation of QuizArena, a web-based gamified multiplayer educational quiz battle system integrating Retrieval-Augmented Generation (RAG), Adaptive Randomization, and Large Language Model (LLM) technologies. It outlines the systematic processes, frameworks, and tools used to guide the study from initial conceptualization through deployment, ensuring that each development decision is grounded in established research principles and aligned with the study's objectives. The chapter is organized around a hybrid CRISP-DM and Agile methodology, wherein CRISP-DM governs the iterative lifecycle of the system's AI-driven components and Agile structures the incremental development of the web application across time-boxed sprints. Together, these frameworks provided a disciplined yet adaptive workflow that enabled the research team to simultaneously advance algorithmic accuracy, software functionality, and user experience quality throughout the project's lifecycle.

Research Design

In this research, a multifaceted research design has been adopted, encompassing developmental, experimental, and descriptive-evaluative research methodologies. Developmental research serves as the primary design framework, guiding the systematic construction, iteration, and refinement of the

QuizArena platform—a web-based gamified multiplayer educational quiz battle system—from initial conceptualization through full deployment. Experimental research is applied in the rigorous evaluation of the core AI algorithms embedded within the system, particularly in measuring the performance of the Retrieval-Augmented Generation (RAG) pipeline using the Retrieval-Augmented Generation Assessment Scale (RAGAS) for Contextual Precision and Answer Relevancy, and in validating the Adaptive Randomization algorithm using Demographic Parity and Bounded Individual Loss metrics for competitive matchmaking fairness. Additionally, descriptive-evaluative research enables a comprehensive assessment of the fully developed system, employing the ISO/IEC 25010 standard to evaluate Functional Suitability, Performance Efficiency, Usability, and Reliability as experienced by student and faculty respondents at the University of Makati. This tri-method research design bridges theory and practice by ensuring that the QuizArena platform is simultaneously developed with technical rigor, validated through objective algorithmic benchmarking, and appraised through standardized user-centric evaluation criteria.

Research Methodology

CRISP-DM and Agile Methodology

For the development of QuizArena, a hybrid CRISP-DM and Agile methodology was adopted as the guiding research and development framework. The Cross-Industry Standard Process for Data Mining (CRISP-DM) was applied to govern the iterative lifecycle of the system's AI-driven components—namely, the Retrieval-Augmented Generation (RAG) pipeline, the Adaptive Randomization algorithm, and the LLM-powered Solution Analyzer—ensuring that each data science task from business understanding through model deployment is carried out systematically and with scientific rigor. Concurrently, the Agile methodology was applied to manage the iterative software development of the QuizArena web application, structuring work into time-boxed sprints that allow continuous integration of user feedback, progressive feature development, and adaptive response to evolving requirements. The synergy between CRISP-DM and Agile is particularly well-suited for this project because QuizArena is simultaneously a data-intensive AI system and a complex real-time web application; both methodologies' shared emphasis on iterative refinement, stakeholder collaboration, and feedback-driven improvement creates a unified development environment where AI model quality and software functionality evolve in parallel. The phases that follow describe how each stage of this combined framework was applied throughout the development of the study.

Plan

The CRISP-DM and Agile methodology commenced with a comprehensive planning phase that aligned data science objectives with software development goals under a unified project roadmap. In the CRISP-DM Business Understanding stage, the research team engaged in extensive domain analysis to clearly define the core business problem: the inadequacy of existing automated assessment platforms in evaluating multi-step reasoning, handwritten mathematical logic, and programming logic within a collaborative, gamified environment. This analysis led to the formulation of precise data science objectives, including the design and evaluation of a RAG pipeline for curriculum-aligned question generation, a covariate-adaptive matchmaking algorithm for equitable team balancing, and an LLM-powered solution analyzer for step-by-step assessment. Key Performance Indicators (KPIs) were established, including RAGAS Contextual Precision and Answer Relevancy scores above 70%, Demographic Parity differences below 0.05, and ISO/IEC 25010 compliance ratings of at least “Agree” across all evaluation dimensions.

Simultaneously, the Agile Sprint Planning process was initiated by constructing an initial product backlog capturing all functional and non-functional requirements of the QuizArena system. Stakeholders were identified as comprising faculty members, students, and the development team within the College of Computing and Information Sciences at the University of Makati. User stories were formulated for each major system feature, including the professor-validated question generation interface, the multiplayer quiz battle room management, the dual-mode scoring system (Speed Mode and Discussion Mode), the designated leader tie-breaker mechanic, the real-time analytics dashboard, and the LLM-driven solution analyzer. The product backlog was prioritized by a combination of business value, technical dependency, and implementation complexity, enabling a logical sprint sequence. A high-level development timeline spanning six sprints of approximately two weeks each was established, with built-in flexibility to accommodate evolving requirements and findings from algorithm validation activities. Version control was established using Git with a GitHub repository, and task tracking was managed through a project board to ensure consistent sprint monitoring and cross-team transparency throughout the development lifecycle.

Design

Following the planning phase, the CRISP-DM and Agile framework guided the comprehensive design of both the system architecture and the algorithmic models underpinning QuizArena's core functionalities. In alignment with CRISP-DM's Data Understanding and Data Preparation stages, the research team conducted an initial data characterization study to identify and understand the structure of the instructional materials that would feed the RAG pipeline. Unstructured PDF syllabi and lecture documents from the College of Computing and Information Sciences were collected and examined to characterize their typographic density, semantic structure, and subject domain distribution. This analysis directly informed critical design decisions regarding the chunking strategy (512-token semantic chunks with 50-token overlap), the embedding model selection (sentence-transformers/all-MiniLM-L6-v2), and the vector store architecture (PGVector integrated into the PostgreSQL database), ensuring that the RAG pipeline design was grounded in the actual characteristics of the data it would process.

Simultaneously, the Agile Design Phase formalized the system architecture of the QuizArena web application. The architecture follows a microservices pattern, separating the Next.js frontend, the Python FastAPI AI microservices backend, and the PostgreSQL database into independently deployable components. The Next.js application handles routing, user session management, and the React-based UI rendered with Tailwind CSS, while the Python microservices host the RAG pipeline via LangChain, the LLM Solution Analyzer, and the Adaptive Randomization algorithm. Real-time game state synchronization between clients is managed through Socket.IO, with Redis serving as an in-memory pub/sub broker to enable horizontal scaling of WebSocket connections. The professor dashboard and student quiz interfaces were designed iteratively through high-fidelity prototypes developed in Figma, validated against user stories and acceptance criteria defined in the product backlog.

The system design artifacts produced in this phase include a Use Case Diagram illustrating the interactions between Professor users and Student users with the platform's major functional modules; an Activity Diagram depicting the lifecycle of a multiplayer quiz session from room creation through result publication; a Context Flow Diagram showing the system's interaction with external entities such as LLM providers and WebSocket clients; a Data Flow Diagram (DFD) detailing the passage of data from professor-uploaded syllabi through the RAG pipeline to the quiz question database; an Entity Relationship

Diagram (ERD) defining the relational schema across the primary entities of Users, Sections, Questions, Sessions, Teams, and Results; and a Data Dictionary formalizing attribute definitions, data types, and constraints for each database entity. These system design artifacts collectively served as the authoritative blueprint for all subsequent development activities.

Develop

The development phase of the CRISP-DM and Agile framework constituted the most extensive stage of the QuizArena project, encompassing the parallel implementation of the AI algorithmic components and the full-stack web application across six iterative sprints. In alignment with the CRISP-DM Modeling and Data Preparation phases, algorithm development proceeded systematically. The RAG pipeline was implemented using LangChain, beginning with an ingestion module that reads professor-uploaded PDF syllabi and lecture documents, applies recursive character text splitting to produce semantically coherent 512-token chunks with 50-token overlap, and embeds them using the sentence-transformers/all-MiniLM-L6-v2 model. Embedded vectors are persisted in a PGVector-enabled PostgreSQL database, enabling cosine similarity search at query time. The retrieval phase constructs a prompt by concatenating the top-k retrieved document chunks with a structured question-generation instruction, which is then submitted to the OpenAI GPT-4o API through LangChain's chat model abstraction. Generated questions are returned in structured JSON format and staged for professor review and validation before being committed to the question bank.

The Adaptive Randomization algorithm for team matchmaking was implemented in Python as a standalone FastAPI microservice. The algorithm ingests each student's historical accuracy score and subject-specific performance covariates, then applies a covariate-adaptive balancing procedure to assign players to teams such that the mean accuracy difference between any two competing teams is minimized. The algorithm was implemented to satisfy the Bounded Individual Loss constraint, ensuring that no individual student is systematically disadvantaged by the matching process across repeated sessions. The LLM Solution Analyzer was implemented as a vision-capable module using GPT-4o's multi-modal API, which accepts base64-encoded images of handwritten mathematical solutions or code screenshots, processes them through a structured prompt requesting step-by-step evaluation against a provided rubric, and returns a structured JSON response containing a per-step accuracy score, corrective feedback, and an overall grade.

The application development followed Agile sprint cycles, with each sprint delivering a functional increment. Early sprints established the authentication system (professor and student roles), section management (section creation, student join requests, and reassignment), and the core question bank interface with manual editing tools. Middle sprints implemented the multiplayer quiz room system with Socket.IO-based real-time state synchronization, including room creation, player joining, team assignment via the Adaptive Randomization algorithm, and the Speed Mode and Discussion Mode scoring logic with the designated leader tie-breaker mechanic. Later sprints integrated the RAG-based question generation endpoint, the Solution Analyzer submission and grading workflow, and the Professor Dashboard featuring real-time match analytics, sortable result views, and comprehensive engagement logs. Throughout all sprints, development followed a test-driven approach with unit tests written using pytest for the Python microservices and Jest for the Next.js frontend, ensuring that each new increment preserved the correctness of previously implemented functionality.

Test

The testing phase of the QuizArena development lifecycle was structured in alignment with the CRISP-DM Evaluation stage and Agile testing principles, encompassing both algorithmic model evaluation and comprehensive system testing to validate the correctness, performance, and usability of the integrated platform. For algorithm evaluation, the RAG pipeline was assessed using the RAGAS framework, computing two primary metrics: Contextual Precision, which measures the proportion of retrieved document chunks that are genuinely relevant to the generated question, and Answer Relevancy, which quantifies the semantic alignment between generated questions and the source instructional content. A curated evaluation dataset of 100 professor-validated question-context pairs was constructed from the University of Makati’s Mathematics and Computer

Science syllabi to serve as the ground truth benchmark. The Adaptive Randomization algorithm was evaluated using Demographic Parity—measuring the difference in team assignment rates across student performance quintiles—and Bounded Individual Loss, which caps the maximum disadvantage any individual student may suffer from matchmaking over repeated sessions. The LLM Solution Analyzer was evaluated on a dataset of 80 annotated handwritten mathematical solution images and 80 structured programming code snippets, computing Weighted F1-Score and inter-rater agreement (Cohen's Kappa) between LLM-assigned scores and human-graded benchmarks.

For system testing, functional testing was conducted through structured test cases derived directly from the acceptance criteria of each user story in the product backlog, validating AI question generation accuracy, solution analyzer behavior, multiplayer state synchronization across concurrent WebSocket connections, and professor editing tool correctness. Integration testing verified the seamless interoperability of the Next.js frontend, the Python FastAPI AI microservices, the PostgreSQL database, the Socket.IO real-time engine, and the Redis pub/sub layer. Non-functional testing encompassed performance testing using Apache JMeter to simulate concurrent user loads representative of a full classroom scenario, as well as stress testing to identify system breaking points under extreme concurrent loads. Security testing was conducted to evaluate the robustness of the JWT-based authentication system and the enforcement of professor-controlled access restrictions. User Acceptance Testing

(UAT) was conducted with a cohort of faculty and students from the University of Makati, gathering usability feedback that was subsequently incorporated into interface refinements in the final sprint. The completed system was evaluated against the ISO/IEC 25010 quality characteristics—Functional Suitability, Performance Efficiency, Usability, and Reliability—using a structured Likert-scale survey instrument administered to UAT participants.

Deploy

Following the successful completion of all testing milestones and the resolution of issues identified during User Acceptance Testing, QuizArena proceeded to the deployment phase, which entailed the configuration, packaging, and release of the fully integrated system for use by the target population at the University of Makati. The Next.js web application was deployed to Vercel, a cloud-native hosting platform optimized for Next.js workloads, with automatic SSL certificate provisioning and global Content Delivery Network (CDN) distribution to ensure low-latency access for on-campus and remote users alike. The Python FastAPI AI microservices, including the RAG pipeline service and the LLM Solution Analyzer service, were containerized using Docker and deployed to a cloud server instance, ensuring isolation, reproducibility, and ease of environment configuration across development, staging, and production environments.

The PostgreSQL database, including the PGVector extension for vector similarity search, was provisioned on a managed database service with automated daily backups and point-in-time recovery capabilities to safeguard all instructional and assessment data. Redis was deployed as a managed in-memory data store to support high-availability WebSocket pub/sub operations during live quiz sessions. The deployment process incorporated a CI/CD pipeline configured through GitHub Actions, automating the build, test, and deployment workflow on every push to the main branch. Post-deployment smoke testing verified the availability and responsiveness of all core endpoints, including the authentication API, the RAG question generation endpoint, the Solution Analyzer submission endpoint, and the WebSocket handshake for multiplayer room management. Monitoring was established using application-level logging to track API response times, WebSocket connection health, and database query performance, enabling proactive identification and resolution of any post-launch performance degradation. End-user documentation comprising an installation guide, a professor user manual, and a student quick-start guide was prepared and made accessible within the platform to support onboarding for all user roles.

Review

The review phase of the CRISP-DM and Agile framework provided a structured mechanism for reflective evaluation and iterative improvement throughout the QuizArena development lifecycle. At the conclusion of each sprint, the development team conducted Sprint Review meetings in which completed features were demonstrated to faculty stakeholders and student representatives from the University of Makati. These review sessions generated actionable feedback that directly shaped subsequent sprint priorities; for example, early sprint reviews led to the redesign of the professor question editing interface to include inline preview functionality, while mid-cycle reviews prompted the addition of a session replay feature to the Professor Dashboard based on faculty feedback regarding post-match analysis needs.

Concurrently, Sprint Retrospective meetings were held after each review to assess team processes, communication effectiveness, and technical practices. Retrospective insights led to the adoption of stricter type annotation conventions in the Python microservices codebase to reduce integration bugs, the introduction of automated RAGAS evaluation scripts into the CI/CD pipeline to detect RAG performance regression across sprints, and the refinement of the WebSocket state reconciliation logic to improve resilience during high-concurrency quiz sessions. In alignment with the CRISP-DM Revisions phase, the RAG pipeline architecture underwent two significant revisions during the project lifecycle: the first revised the document chunking strategy from fixed-length to recursive character-based splitting after initial RAGAS evaluation indicated suboptimal contextual coherence, and the second introduced a re-ranking step using cross-encoder models to improve the precision of retrieved document chunks prior to question generation. Each of these revisions was managed within the Agile sprint framework, ensuring that improvements to the AI components were delivered incrementally alongside continued progress on the application software. The review phase ultimately ensured that QuizArena was not merely functionally complete at deployment, but continuously refined and validated against both algorithmic benchmarks and real-world stakeholder expectations throughout its entire development lifecycle.

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